

Modeling and Analysis of a New Cylindrical Magnetic Levitation Gravity Compensator With Low Stiffness for the 6-DOF Fine Stage

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Abstract—A novel cylindrical magnetic levitation gravity compensator (MLGC) with low-stiffness and vacuum-compatible characteristics is proposed in this paper. This compensator can be used as the out-of-plane electromagnetic actuator for the 6-DOF fine stage in certain high-precision positioning applications, for example, the wafer stage in a lithography machine. Compared with conventional actuators such as the electromagnet and voice coil motor, the heat and the resulting temperature rise that degrade the stage positioning accuracy can be reduced by using passive magnetic gravity compensation. Based on the equivalent current method, the analytical equations for the magnetic field, static levitation force, vertical stiffness, and dynamic levitation force are derived. However, the static levitation force–vertical displacement characteristic from the traditional analytical model is not sufficiently accurate for the low-stiffness applications when compared with the finite-element model. Therefore, the main reason for the model error is analyzed, and an improved semianalytical method based on a single-point magnetostatic field simulation is proposed. This method offers a theoretical basis for the analysis and design of the low-stiffness MLGC.

Index Terms—Electromagnetic modeling, gravity compensator, low stiffness, magnetic levitation, vibration isolation.

I. INTRODUCTION

PRECISION positioning in multi-degrees-of-freedom is usually required in many advanced industrial applications, such as microscopy, biomedical science, and semiconductor lithography [1]–[8]. At present, the combination of a coarse stage and a fine stage is a widely adopted scheme for the high-precision positioning wafer stage, which can simultaneously satisfy the requirements of a long stroke and high precision. The coarse stage is used to achieve a large-stroke 2-DOF motion (hundreds of millimeters in the xy plane) with micron-level accuracy, and the fine stage is used to achieve a short-stroke 6-DOF motion (a few millimeters) with nano-level accuracy. In recent decades, the technical requirements of the 6-DOF fine stage have been gradually upgraded with the improvements of lithography resolution.

Manuscript received April 4, 2014; revised June 19, 2014 and September 5, 2014; accepted October 11, 2014. Date of publication October 29, 2014; date of current version May 8, 2015.

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Digital Object Identifier 10.1109/TIE.2014.2365754

In general, the 6-DOF motion can be divided into in-plane motion (x, y, θ_z) and out-of-plane motion (z, θ_x, θ_y) . For the latter, a combination scheme involving several z actuators with a symmetrical distribution is often adopted, in which the z actuator is an electromagnet [9], [10] or a voice coil motor [11]–[13]. The vertical displacement and pitch angle can be achieved by using the reluctance force or the Lorentz force. However, continuous power loss is inevitable for these two types of z actuators to support a large moving mass during the positioning process. The amount of heat produced by the coils will lead not only to an environmental temperature rise but also to unwanted mechanical deformation. Both of these influences will result in a decrease in positioning accuracy. Therefore, the concept of gravity compensation is presented. In order to compensate for the moving mass, mechanical springs, air bearings, and passive magnetic levitation have been discussed. Among these methods, the mechanical spring is subject to mechanical coupling, which limits the positioning accuracy of the fine stage. Air bearing is a good choice for gravity compensation because of its stable characteristics and mature technologies. However, the system structure will be very complex if air bearings are used in next-generation lithography (extreme ultraviolet lithography), for which a high vacuum condition is required. In contrast, passive magnetic levitation has many advantages, such as no mechanical contact, high force density, vacuum compatibility, and structure simplicity.

Related research projects on passive magnetic levitation and gravity compensation have been discussed in a number of papers. In [14], the coupling energy, force, and stiffness of different configurations of magnetic bearings and magnetic couplings were approximately calculated. The model was based on the assumption that the gap between the magnets was very small in comparison with the radius. A semianalytical calculation of the interaction between the ring-shaped permanent magnets was presented in [15]. In [16], the analytical model of the magnetic force for the axial stack permanent-magnet bearings was established and verified by the ANSYS simulation. A ring-shaped passive vibration isolator was proposed and the vibration isolation performance of the isolator was tested in [17]. A new magnetic gravity compensation system using permanent magnets and a Lorentz coil was proposed in [18]. In [19] and [20], a flat-type active magnetic bearing using Halbach magnet arrays and Lorentz coils for a magnetic levitation stage was proposed. Large forces and low stiffness could be achieved through design optimization. In [21], a magnetic levitation table supported by repulsive forces of four sets of permanent magnets

was researched. A vibration isolation system combining zero-power magnetic suspension with springs, which yielded a good performance of vibration isolation, was proposed in [22]. In [23], the design criteria and parameter optimization of a quasi-zero stiffness magnetic levitation system for vibration isolation were presented. An electromagnetic isolator for gravity compensation of a heavy payload by passive permanent magnetic force and active Lorentz force was designed in [24]. In [25], improved 3-D analytical expression for the torque between the permanent magnets in magnetic bearings was derived by using the Lorentz force method. In [26], unequal magnet arrays were adopted in a magnetic gravity compensator to realize the goal of a large passive force and low stiffness.

One of the major problems with passive magnetic levitation using permanent magnets is the position-dependent feature. The magnetic field of a permanent magnet exponentially decreases with an increase in the relative distance. Therefore, the stiffness of the magnetic levitation, which is defined as the rate of the force variation with displacement, is large. For the fine stage, it is important to prevent the vibration transmission from the coarse stage to the upper silicon wafer in the process of lithography. Therefore, the stiffness of the passive magnetic levitation should be kept as low as possible.

In this paper, a novel cylindrical magnetic levitation gravity compensator (MLGC) is proposed, which can be used as the z actuator for the fine stage. The MLGC provides an integrated solution involving passive gravity compensation and active positioning control by employing passive magnetic levitation embedded with Lorentz coils. Because the Halbach magnet array is applied in the mover, the levitation force is large, and the stiffness is low compared with the non-Halbach structure. Although the Halbach magnet array is widely adopted in many types of motors and actuators [27]–[31], a special problem will occur when it is applied in the vibration isolation application, for which low stiffness is needed. The problem is that the zero stiffness position drifts when the stiffness decreases to a certain value, which has a negative effect on the vibration isolation performance. However, to date, this problem has not been well researched. The primary contribution of this paper is to analyze the leading cause of the mentioned problem and to propose an improved semianalytical model. Through finite-element simulation and experiment, the feasibility of the proposed model is effectively verified.

This paper is organized as follows. In Section II, the structure and principle of the MLGC are proposed. In Section III, the traditional analytical model for the MLGC is established. In Section IV, the inaccuracy of the analytical model is noted by comparing it with the finite-element model. Then, we propose an improved semianalytical model in which the actual difference in the operating points of the stator magnet is considered. In Section V, the relevant experimental verification is completed.

II. BASIC CONFIGURATION OF THE MLGC

Fig. 1 shows the cross-sectional configuration of the MLGC. The stator is composed of one permanent magnet ring with radial magnetization, four Lorentz coils, the coils frame, and

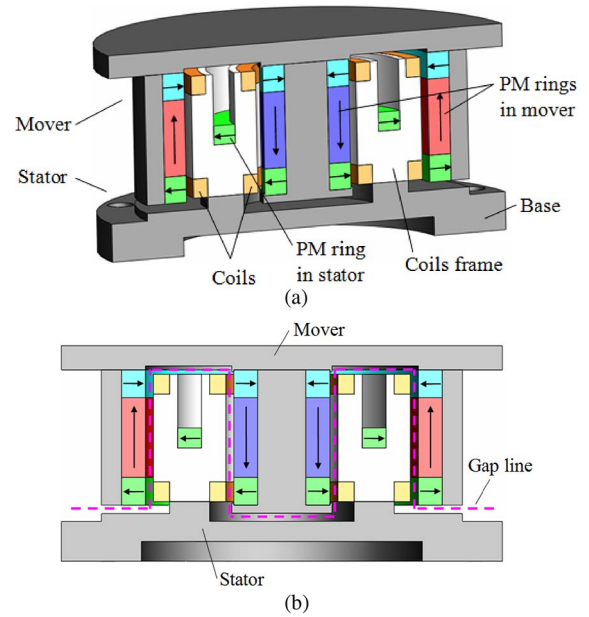


Fig. 1. Cross-sectional configuration of the MLGC. (a) 3-D view. (b) Side view.

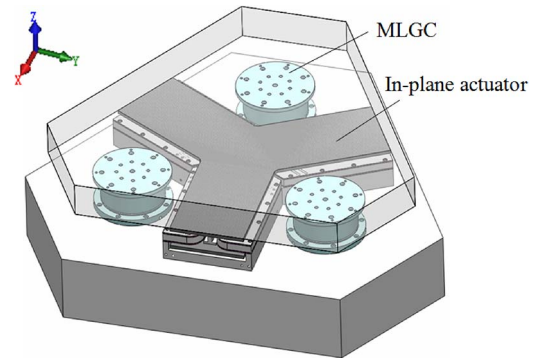


Fig. 2. Illustration of the fine stage driven by a short-stroke planar motor and three MLGCs.

the base. The mover is composed of six permanent magnet rings and the magnets frame. The six permanent magnet rings can be divided into two groups of Halbach magnet arrays, i.e., the inner ring and the outer ring. Therefore, each group consists of three permanent magnet rings, which have 90° differences in their direction of magnetization.

The initial position of the MLGC is defined when the mover magnet rings are symmetrically distributed with respect to the stator magnet ring, as shown in Fig. 1. According to the objective law of repelling the same polarity and attracting the opposite polarity, the upper magnet rings of the mover repulse the stator ring, and the lower magnet rings of the mover attract the stator ring. Therefore, a static levitation force will be generated in the mover, which can be used to compensate for the moving mass of the fine stage. At the same time, four Lorentz coils are embedded in the stator to realize the dynamic control in the levitation direction. The four Lorentz coils are close to the four magnet rings with radial magnetization. The radial component of the magnetic field interacts with the circumferential current, by which the dynamic levitation force will be produced.

Generally, the fine stage will be levitated and positioned in the z -direction using three or four MLGCs with a symmetrical

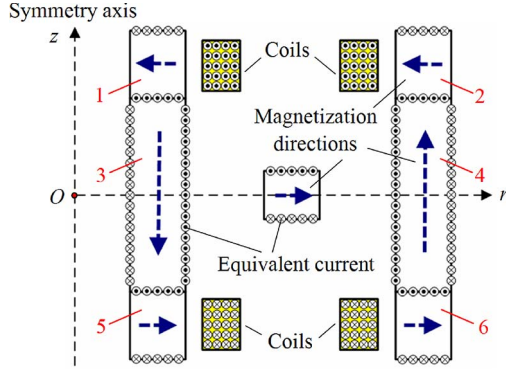


Fig. 3. Equivalent surface current model of the MLGC.

arrangement, as shown in Fig. 2. Thus, each MLGC is required to compensate for one third or one quarter of the moving mass. Additionally, the effective motion ranges of the MLGC for three axes are all ± 1 mm. The MLGC has three main functions. The first function is to compensate for the moving mass passively and to reduce the copper loss. The second function is the active control of positioning in the z -direction. The third function is to suppress the vibration transmission from the coarse stage to the fine stage. For in-plane motion, the short-stroke planar motor with high precision [32]–[34] is commonly used as the in-plane actuator, but this content is not discussed in this paper.

III. TRADITIONAL ANALYTICAL MODEL OF THE MLGC BASED ON THE EQUIVALENT CURRENT METHOD

In [27], Trumper *et al.* provided an equivalent current model for the Halbach array and a general analysis framework for linear permanent-magnet machines. Here, the model is applied to the MLGC. To calculate the force and stiffness, the air gap magnetic field should be obtained first.

A. Magnetic Field

Fig. 3 shows the equivalent surface current model of the MLGC. The permanent magnet ring with radial magnetization is represented by two identical current sheets whose normal vectors are parallel to the z -axis. In addition, the permanent magnet ring with axial magnetization is represented by two concentric current sheets with different radii whose normal vectors are perpendicular to the z -axis. The magnetic vector potential in any field point produced by the surface current density is expressed as

$$\vec{A}(\vec{e}) = \frac{\mu_0}{4\pi} \oint_S \frac{\vec{j}_m(\vec{e}')}{|\vec{e} - \vec{e}'|} ds' \quad (1)$$

where μ_0 is the vacuum permeability, $\vec{j}_m$ is the surface current density, $\vec{e}'$ is the point on the surface current sheet, and $\vec{e}$ is an arbitrary field point.

The equivalent surface current density is expressed as

$$\vec{j}_m = \vec{M} \times \vec{n} \quad (2)$$

where M is the magnetization of the permanent magnet, and $\vec{n}$ is the normal unit vector.

The magnetic flux density is expressed as

$$\vec{B} = \nabla \times \vec{A}. \quad (3)$$

By inserting (1) and (2) into (3), the magnetic flux density can be derived. When we calculate the levitation force of the MLGC, only the radial component of the magnetic flux density is needed, which is given as

$$B_{r,\text{radial}} = \frac{\mu_0 M}{4\pi} \sum_{k=1}^2 (-1)^{(k+1)} \cdot \int_{r(1)}^{r(2)} \int_0^{2\pi} \frac{(z - z(k)) \cos(\phi') r' \cdot d\phi' dr'}{\left[r^2 + r'^2 - 2rr' \cos(\phi') + (z - z(k))^2 \right]^{\frac{3}{2}}} \quad (4)$$

$$B_{r,\text{axial}} = \frac{\mu_0 M}{4\pi} \sum_{j=1}^2 \sum_{k=1}^2 (-1)^{(j+k)} \cdot \int_0^{2\pi} \frac{\cos(\phi') \cdot d\phi'}{\left[r^2 + r(j)^2 - 2rr(j) \cos(\phi') + (z - z(k))^2 \right]^{\frac{1}{2}}} \quad (5)$$

where $B_{r,\text{radial}}$ and $B_{r,\text{axial}}$ are the radial components of the magnetic flux density from the magnet ring with radial magnetization and axial magnetization, respectively; $r(1)$ and $r(2)$ are the inner radius and the outer radius of a magnet ring, respectively; and $z(1)$ and $z(2)$ are the lower face and the upper face of a magnet ring, respectively.

The total air gap magnetic field can be obtained by superposing the contribution of all the permanent magnet rings. The expression is

$$B_r = \sum_{i=1,2,5,6} B_{ri,\text{radial}} + \sum_{j=3,4} B_{rj,\text{axial}}. \quad (6)$$

B. Static Levitation Force and Vertical Stiffness

The static levitation force is the interaction between the stator magnet ring and the mover magnet rings. Because the stator magnet ring can be also represented by two current sheets, a vertical force will be generated by the interaction between the air gap magnetic field and the current sheets. According to Lorentz's force law

$$\vec{F} = \iiint_V \vec{J} \times \vec{B} \cdot dV. \quad (7)$$

The static levitation force is given as

$$F_{z,\text{static}}(z_{ss}) = \int_0^{2\pi} \int_{R_i}^{R_o} B_r \left(r, \frac{h_m}{2} + z_{ss} \right) (-J) r dr d\phi \\ + \int_0^{2\pi} \int_{R_i}^{R_o} B_r \left(r, -\frac{h_m}{2} + z_{ss} \right) (J) r dr d\phi$$

$$= 2\pi J \int_{R_i}^{R_o} \left[B_r \left(r, -\frac{h_m}{2} + z_{ss} \right) - B_r \left(r, \frac{h_m}{2} + z_{ss} \right) \right] r dr \quad (8)$$

where z_{ss} is the displacement in the levitation direction, h_m is the thickness of the stator magnet ring, R_i is the inner radius of the stator magnet ring, and R_o is the outer radius of the stator magnet ring.

The vertical stiffness between the stator and mover magnet rings is calculated by

$$k_z(z_{ss}) = \frac{\partial F_{z,static}}{\partial z_{ss}}. \quad (9)$$

C. Dynamic Levitation Force

There are four coils in the cylindrical MLGC, and the four coils are symmetrically distributed around the four permanent magnet rings with radial magnetization. Under the radial magnetic field, a dynamic levitation force can be produced by the coils. To simplify the model, each coil is modeled as a number of current sheets. The dynamic force calculation of each current sheet is the same as the static force calculation. Finally, the dynamic levitation force is the superposition of all the current sheet forces. The expression for the dynamic levitation force is given as

$$F_{z,dynamic}(z_{ss}) = F_{z,dynamic1}(z_{ss}) + F_{z,dynamic2}(z_{ss}) + F_{z,dynamic3}(z_{ss}) + F_{z,dynamic4}(z_{ss}). \quad (10)$$

IV. MODEL IMPROVEMENT BASED ON A SINGLE-POINT MAGNETOSTATIC FIELD SIMULATION

A. Comparison Between the Traditional Analytical Model and the Finite-Element Model

To verify the analytical model above, the finite-element model of the MLGC is established using commercial finite-element software—ANSYS Maxwell [35]. Because of the axial symmetry property of the MLGC, only a 2-D finite-element model is required. The 2-D finite-element model of the MLGC is shown in Fig. 4. The requirements for the MLGC are given in Table I. Through structural parameter optimization, the initial dimensions of the MLGC can be determined and are shown in Table II. In the finite-element model, all of the permanent magnet rings are NdFeB35 with $H_{cb} = -890\,000$ A/m, $B_{rem} = 1.23$ T, and $\mu_r = 1.0998$.

Fig. 5 shows the static levitation force–vertical displacement characteristic curves obtained from the analytical model and the finite-element model. The curve in Fig. 5(a) is symmetrical with respect to the initial position, and the force reaches a maximum value at $z_{ss} = 0$. In contrast, the curve in Fig. 5(b) is a decreasing function of the vertical displacement. The average error between the two models is approximately 11.55%. Through quadratic polynomial fitting with least square method,

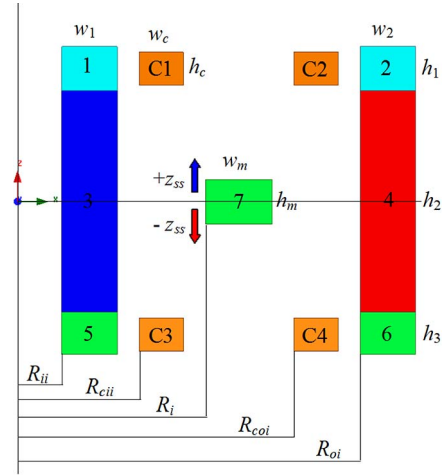


Fig. 4. Two-dimensional finite-element model of the MLGC.

TABLE I
REQUIREMENTS FOR THE MLGC

Item	Symbol	Value	Unit
Static levitation force	$F_{z,static}/F_{zs}$	40	N
Dynamic levitation force	$F_{z,dynamic}/F_{zd}$	± 4	N
Motion range	(x_{ss}, y_{ss}, z_{ss})	$(\pm 1, \pm 1, \pm 1)$	mm
Vertical stiffness	K_z	< 100	N/m
Mover mass	m_{mover}	0.5	kg
Stator mass	m_{stator}	0.5	kg
Current density of coils	J_c	5	A/mm ²
Overall dimension restriction	$D \cdot H$	$D95 \cdot H60$	mm

TABLE II
GEOMETRIC PARAMETERS OF THE FINITE-ELEMENT MODEL

Item	Symbol	Value	Unit
mover	inner radius of inner magnet ring	R_{ii}	4 mm
	inner radius of outer magnet ring	R_{oi}	31 mm
	thickness of upper magnet ring	h_1	4 mm
	thickness of middle magnet ring	h_2	20 mm
	thickness of lower magnet ring	h_3	4 mm
	width of inner magnet ring	w_1	5 mm
	width of outer magnet ring	w_2	5 mm
stator	inner radius of inner coil	R_{cii}	11 mm
	inner radius of stator ring	R_i	17 mm
	inner radius of outer coil	R_{coi}	25 mm
	thickness of stator ring	h_m	4 mm
	thickness of coil	h_c	3 mm
	width of stator ring	w_m	6 mm
	width of coil	w_c	4 mm

a continuous function for the static levitation force is obtained. Based on (9), the stiffness can be calculated. Fig. 6 shows the vertical stiffness–vertical displacement characteristic curves. Because of the symmetry of force, there is a zero stiffness position at $z_{ss} = 0$ for the analytical model. In contrast, most of the vertical stiffness is negative within the range of $z_{ss} = [-1, 1]$ mm for the finite-element model.

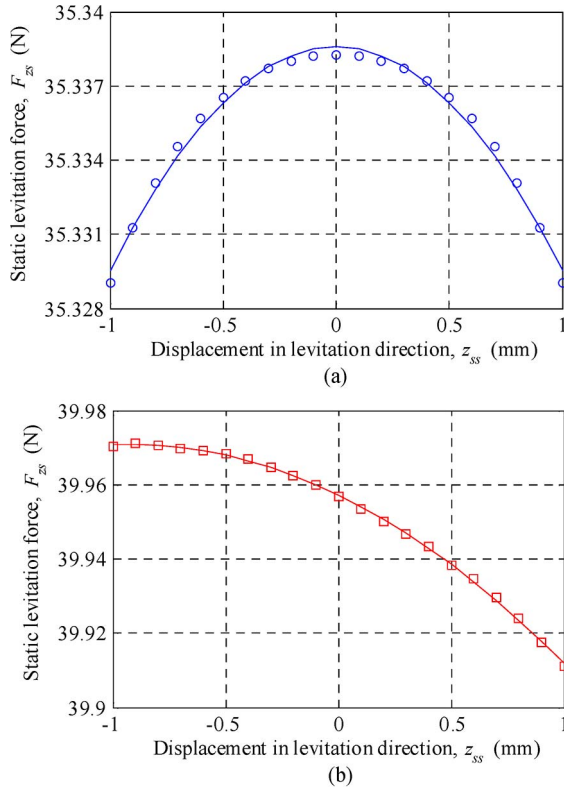


Fig. 5. Static levitation force of the MLGC as a function of the vertical displacement. (a) Analytical result. (b) Finite-element result.

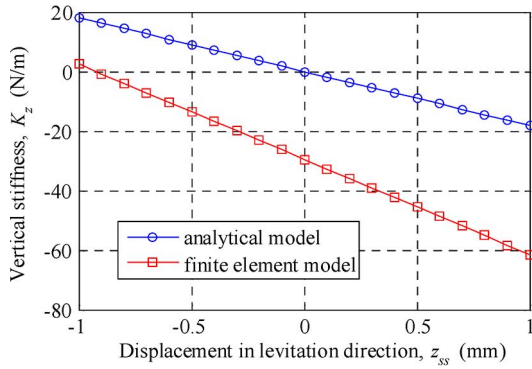


Fig. 6. Comparison of the vertical stiffness of the MLGC as a function of the vertical displacement.

In conclusion, the analytical model does not match well with the finite-element model. First, the error for the levitation force is large. Second, the slope of the levitation force versus the displacement curve has completely changed, which will cause a decisive change in the vertical stiffness distribution within the effective vertical displacement. For low-stiffness applications, the problem cannot be neglected and needs to be solved.

B. Reason Analysis for the Model's Inaccuracy

To obtain a simplified analytical model, the relative permeability of the permanent magnet in the traditional analytical model is always supposed to be 1.0, and hence, the magnetization is equal to the coercivity H_{cb} [18]. In fact, the relative permeability $\mu_r > 1.0$, and the actual magnetization is larger

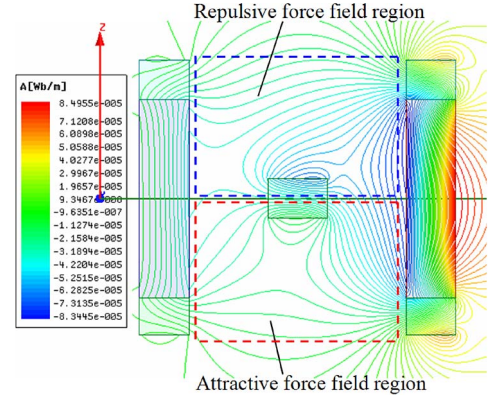


Fig. 7. Magnetic flux distribution of the MLGC (without coils).

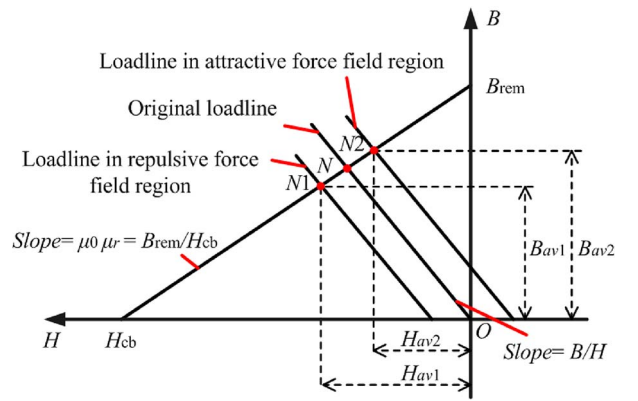


Fig. 8. Schematic that explains the magnetic reaction.

than H_{cb} . This is the main reason for the lower analytical value. From another point of view, the magnetic field is more confined to the magnet volume because of the relative permeability, which is not considered in the traditional model.

As to the change of the slope for the levitation force versus the displacement curve, the reason is that the attractive force is seen entirely equal to the repulsive force. For most applications, this assumption is reasonable and has been widely adopted [36]–[41]. However, this assumption will cause a decisive error for the gravity compensation or vibration isolation in which low stiffness is needed, particularly in the magnetic device in which the attractive force and the repulsive force both exist. For low-stiffness applications, the force variation is very small, and the force becomes highly sensitive to the displacement.

Fig. 7 shows the magnetic flux distribution of the MLGC (without coils). The stator magnet ring is located in the air gap magnetic field generated from the mover Halbach array. Additionally, the stator ring is simultaneously affected by the upper repulsive force and the lower attractive force. Here, the upper half part is defined as the repulsive force field region, and the lower half part is defined as the attractive force field region.

As shown in Fig. 8, the operating point will move up when the magnet is under an attractive force field. In contrast, the operating point will move down when the magnet is under a repulsive force field. Therefore, the attractive force should be slightly larger than the repulsive force between two magnets. Being analogous to the "armature reaction," the change in the

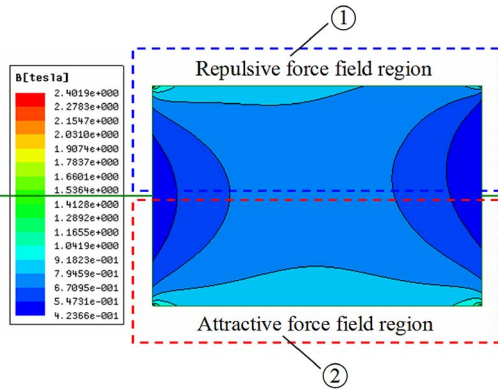


Fig. 9. Magnetic flux density of the stator magnet ring.

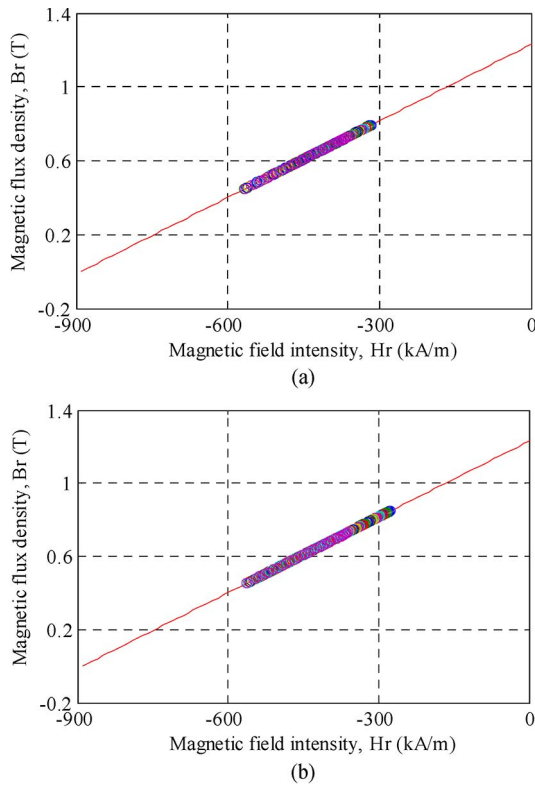


Fig. 10. Operating points of the stator permanent magnet. (a) Points in the repulsive force field region. (b) Points in the attractive force field region.

operating point of a magnet under an external magnetic field can be defined as the “magnetic reaction.”

C. Operating Point of the Stator Permanent Magnet Ring

Fig. 9 shows the magnetic flux density of the stator magnet ring. It can be seen that the magnetic flux density of the lower part in the attractive force field region is higher than that of the upper part in the repulsive force field region because of the magnetic reaction.

The operating point of the magnet is calculated by using the magnetostatic field finite-element simulation. Fig. 10 shows the operating points of the upper half and lower half of the stator magnet. It is observed that the maximum operating point of the lower part is higher than that of the upper part.

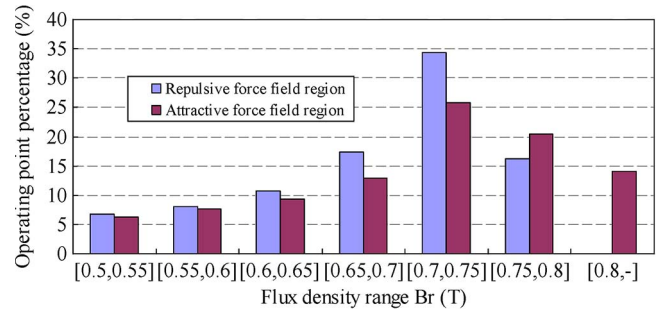


Fig. 11. Comparison of the operating points of the magnet parts in the repulsive force field region and the attractive force field region.

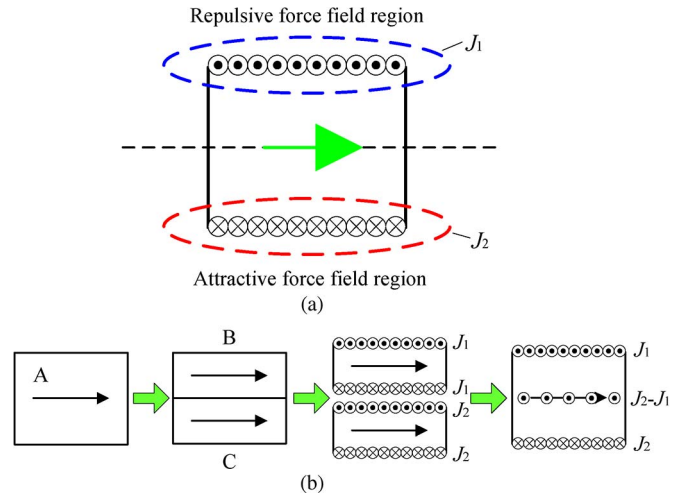


Fig. 12. Improved equivalent surface current model for the stator magnet. (a) Final model. (b) Equivalent process.

Fig. 11 shows a comparison of the operating points for the magnet parts in the repulsive force field region and in the attractive force field region. The major operating points in the repulsive force field region are in the range of $[0.7, 0.75]$ T. In contrast, more operating points in the attractive force field region are located in the range of $[0.75, 0.8]$ and $[0.8, -]$ T.

D. Improved Semianalytical Model

Observing the analytical model, the stator magnet is represented by two current sheets when calculating the static levitation force. The surface current density is equal to the magnetization of the permanent magnet. For the traditional concept, the two equivalent currents are the same. Therefore, the static levitation force–vertical displacement characteristic curve is always exactly symmetrical. From the preceding discussion, the actual operating points of the magnet are needed to be considered for low-stiffness applications. Hence, a semianalytical model based on a single-point magnetostatic field simulation can be used to compensate for the original model error and to accurately calculate the static levitation force–vertical displacement characteristic curve of the MLGC.

Fig. 12(a) shows the improved equivalent surface current model of the stator magnet. The upper current sheet is in the repulsive force field region, and the lower current sheet is in the attractive force field region. In order to distinguish the two

TABLE III
 EQUIVALENT CURRENT DENSITY OF THE STATOR MAGNET OBTAINED
 FROM A SINGLE-POINT SIMULATION

Item	Value	Unit
H_{av1}	404.05	kA/m
H_{av2}	382.32	kA/m
J_1	938.49	kA/m
J_2	940.66	kA/m

current sheets from each other, the former current density is J_1 , and the latter current density is J_2 . Fig. 12(b) shows the equivalent process of the unsymmetric currents for the stator magnet. First, the stator magnet A can be replaced by two uniform-sized magnets B and C. As magnets B and C are located in the repulsive force field region and in the attractive force field region, respectively, the average operating points for the two magnets are different. The one in the attractive force field region is a little higher than the other, i.e., $J_2 > J_1$. The lower surface of B overlaps with the upper surface of C, which forms a middle layer whose current density is $J_2 - J_1$. Therefore, the stator magnet can be represented by three current surfaces. However, the middle layer can be neglected due to the low current density. Finally, the equivalent model of the stator magnet using unsymmetric currents, shown in Fig. 12(a), is built.

According to Fig. 10, the average operating point of the two magnet parts can be obtained. Then, the actual magnetizations, i.e., the current density, can be further derived. The expressions for the two current densities can be expressed as

$$\begin{cases} J_1 = M_1 = \mu_r H_c - (\mu_r - 1) H_{av1} \\ J_2 = M_2 = \mu_r H_c - (\mu_r - 1) H_{av2} \end{cases} \quad (11)$$

where H_{av1} and H_{av2} represent the average magnetic field intensity of the magnet parts in the repulsive force field region and the attractive force field region, respectively.

Moreover, (8) for calculating the static levitation force should be modified as

$$\begin{aligned} F_{z,static}(z_{ss}) &= \int_0^{2\pi} \int_{R_i}^{R_o} B_r \left(r, \frac{h_m}{2} + z_{ss} \right) (-J_1) r dr d\phi \\ &+ \int_0^{2\pi} \int_{R_i}^{R_o} B_r \left(r, -\frac{h_m}{2} + z_{ss} \right) (J_2) r dr d\phi. \end{aligned} \quad (12)$$

Additionally, it must be noticed that the actual operating points of the mover magnets are also required in order to get accurate force and stiffness. Based on the preceding magnetostatic field simulation, the average magnetic field intensity and the average operating point for six mover magnets can be calculated. Next, the air gap magnetic field in (12) should be updated according to the actual magnetizations of six mover magnets. From the simulation results, the current density of the stator magnet and the magnetizations of the mover magnets are calculated, as shown in Tables III and IV, respectively.

 TABLE IV
 ACTUAL OPERATING POINTS FOR THE MOVER MAGNETS OBTAINED
 FROM A SINGLE-POINT SIMULATION

Item	Average Magnetic field intensity (kA/m)	Magnetization (kA/m)
Magnet 1	365.10	942.37
Magnet 2	425.50	936.35
Magnet 3	110.61	967.77
Magnet 4	209.93	957.86
Magnet 5	350.00	943.88
Magnet 6	416.70	937.23

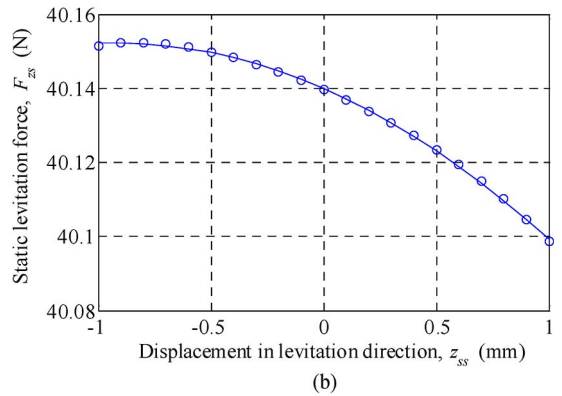
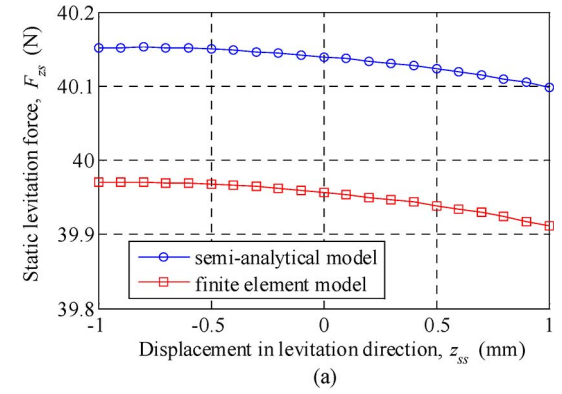


Fig. 13. Static levitation force of the MLGC as a function of the vertical displacement from the semianalytical model. (a) Comparison with the finite-element model. (b) Enlarged curve from the semianalytical model.

Fig. 13 shows the static levitation force–the vertical displacement characteristic curve obtained from the semianalytical model. As shown in Fig. 13(a), the average error of the static levitation force between the semianalytical model and the finite-element model is only 0.46%. Additionally, the variation trend of the force as a function of the displacement can be shown to be correct by comparing Figs. 5(b) and 13(b). Fig. 14 shows a comparison of the vertical stiffness between the two models. It can be seen that the semianalytical result of the stiffness becomes closer to the finite-element result compared with the traditional analytical model.

E. Comparison Between the Proposed Model and the Finite-Element Model

In fact, only exact values of the operating points at one single z position can be obtained by using a single-point magnetostatic

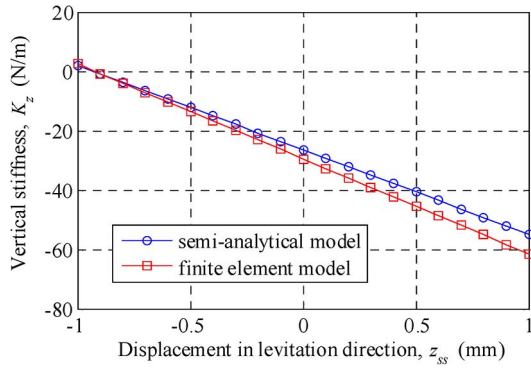


Fig. 14. Vertical stiffness of the MLGC as a function of the vertical displacement.

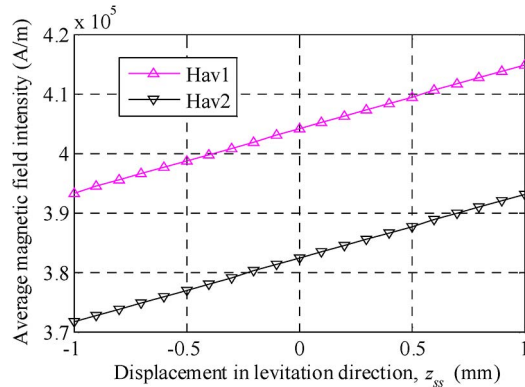


Fig. 15. Average magnetic field intensity of the stator magnet as a function of the vertical displacement from the finite-element model.

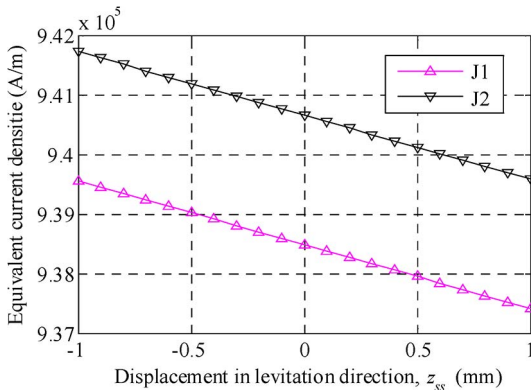


Fig. 16. Equivalent current density of the stator magnet.

field simulation. Additionally, the operating points of the stator magnet will change when the vertical displacement changes. However, the deviation of the average operating points between the upper half and the lower half of the stator magnet remains constant when the vertical displacement changes, as shown in Fig. 15. Therefore, the deviation between J_1 and J_2 within the effective range of motion shown in Fig. 16 can be obtained only by a single-point simulation. The general variation trend of the force–displacement characteristic curve will not change significantly when only the average operating point at $z_{ss} = 0$ is adopted. The feasibility has been verified by using the finite-element model in the previous section.

Generally, the total time for solving a 2-D model in the finite-element software is very short. However, the simulation

time will be greatly increased for low-stiffness applications because of the nearly constant force. To obtain a smooth force–displacement characteristic curve, we need to increase the mesh quantity and decrease the percent error at the same time. These two aspects will cause the computing time to increase significantly. After repeated attempts, the smooth force–displacement curve will appear when the total number of mesh elements rises to 2.2×10^5 and the percent error is reduced to 1×10^{-5} . Then, the total time for obtaining a complete force–displacement curve that contains 21 points from $z_{ss} = -1$ mm to $z_{ss} = 1$ mm will be approximately 8–11 h. It is obvious that the finite-element simulation is not a rapid method for analyzing the MLGC for low-stiffness passive gravity compensation.

In contrast, the computing time for obtaining a complete force–displacement curve for the MLGC using the semianalytical method is apparently shorter than the time required for the finite-element simulation because only the one-point magneto-static field simulation at $z_{ss} = 0$ is required. In addition, the calculation of the average operating points is easily achieved with MATLAB or other data processing software.

F. Adjustment of the Zero Stiffness Point

It is shown in Fig. 14 that the zero stiffness point drifts as a result of the magnetic reaction. Additionally, the direction of the drift tends toward the attractive force field region because the force in the attractive force field region is larger than the force in the repulsive force field region. In Fig. 13(b), the resultant levitation force will continue to rise to a certain value before starting to decrease when the vertical displacement of the stator magnet moves to the attractive force field region because of the larger attractive force. Therefore, the zero stiffness point of the MLGC will appear in the attractive region ($z_{ss} < 0$). To obtain low stiffness for gravity compensation within the range of $z_{ss} = [-1, 1]$ mm, the zero stiffness position needs to be adjusted near to $z_{ss} = 0$ as close as possible. The goal of the adjustment is to weaken the magnetic field in the attractive force field region or to enhance the magnetic field in the repulsive force field region. There are three main adjustment methods. The first method is to readjust the initial relative position between the mover and the stator. The second method is to increase the thickness of the magnetic source of the repulsive force field. The third is to decrease the thickness of the magnetic source of the attractive force field. The third method is adopted in this paper.

V. EXPERIMENT

To verify the semianalytical model and the finite-element model presented above, one prototype for the MLGC was manufactured. Fig. 17 shows the prototype for the MLGC. The geometric parameters of this prototype are the same as the parameters in Table II, except that the parameter h_3 is modified to 3.8 mm. Because a magnet ring with perfect radial magnetization is difficult to manufacture and is very expensive, the tile magnet with parallel magnetization was adopted as an alternative solution. There are five magnet rings with radial magnetization in the MLGC. One of them is in the stator, and

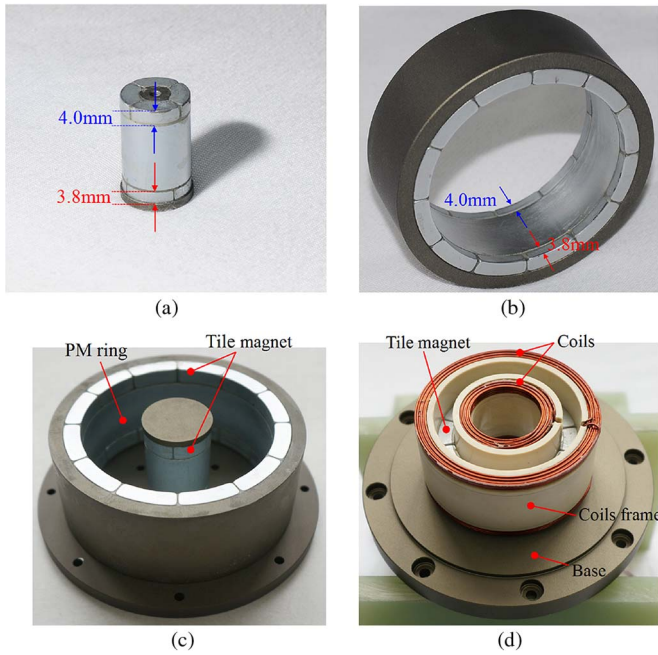


Fig. 17. Prototype of the MLGC. (a) Inner ring of mover. (b) Outer ring of mover. (c) Mover. (d) Stator.

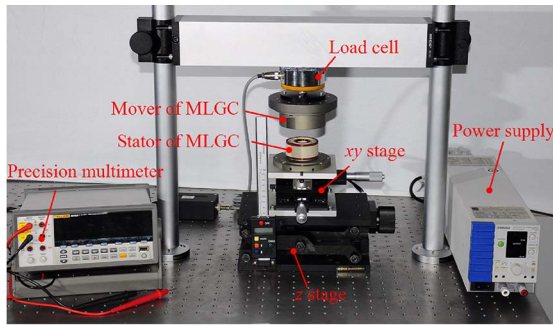


Fig. 18. Experimental setup of the MLGC.

the other four are in the mover. As shown in Fig. 17(a) and (b), the inner magnet ring is composed of four tile magnets, and the outer magnet ring is composed of 12 tile magnets. As shown in Fig. 17(d), the magnet ring in the stator is composed of 12 tile magnets. For the magnet with axial magnetization, a single ring is used. The magnet housing and the base are made of aluminum alloy 7075 with superficial oxide treatment. To eliminate eddy currents, the coil frame is made of a nonmetallic material (PEEK). The masses of the mover and stator are 0.467 and 0.410 kg, respectively.

To evaluate the levitation force characteristics of the prototype, a three-axis test platform is set up, as shown in Fig. 18. The mover of the MLGC is fixed to a beam structure through the load cell. The position of the stator can be changed by the xy stage and the z stage. Fig. 19 shows the static characteristics of the MLGC. The stiffness is obtained from the force–displacement curve by using (9). The measured nominal levitation force is 39.13 N, which matches well with the proposed semianalytical model and the finite-element model. As shown in Fig. 19(b), the zero stiffness point has been confined to the range of $z_{ss} = [0, 0.5]$ mm by adjusting the parameter h_3 .

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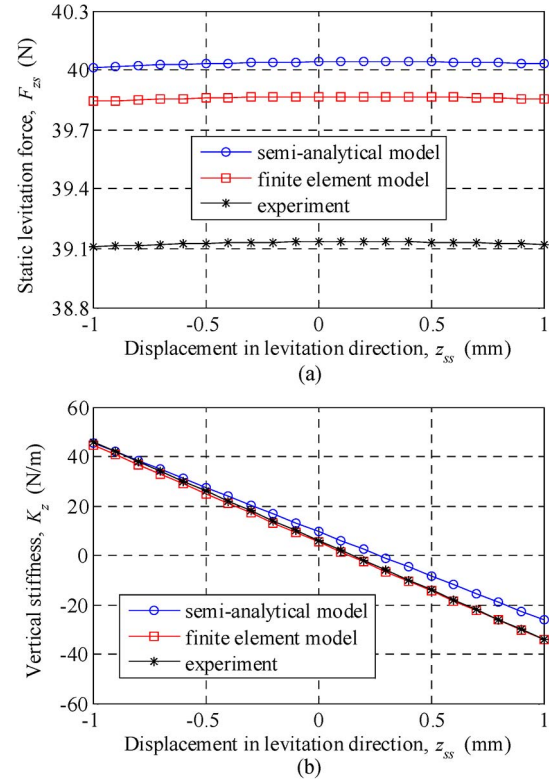


Fig. 19. Comparison of the static characteristics of the MLGC. (a) Static levitation force. (b) Vertical stiffness.

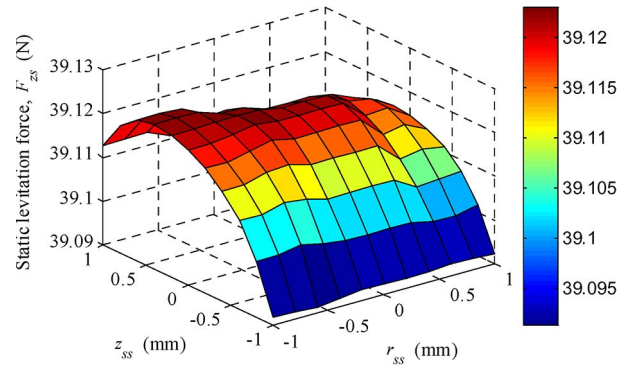


Fig. 20. Static levitation force distribution of the MLGC prototype.

Fig. 20 shows the static levitation force distribution with different vertical displacement (z_{ss}) and radial displacement (r_{ss}). In Fig. 20, the variation of the static levitation force is quite small for the whole plane. Additionally, the force variation with the radial displacement is negligible.

Fig. 21 shows the vertical stiffness distribution calculated from Fig. 20 by quadratic polynomial fitting and derivation. It can be seen that the vertical stiffness is low within the effective range of motion. The maximum negative vertical stiffness of the prototype is $K_z = -43.64$ N/m. The maximum positive vertical stiffness is $K_z = 53.44$ N/m.

Fig. 22 shows the resultant levitation force distribution of the MLGC prototype. The resultant levitation force is the superposition of the static force and the dynamic force. When a current of ± 1.6 A is applied to the four coils, a nominal

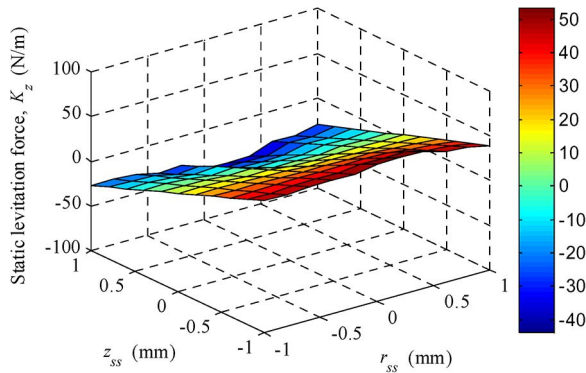


Fig. 21. Vertical stiffness distribution of the MLGC prototype.

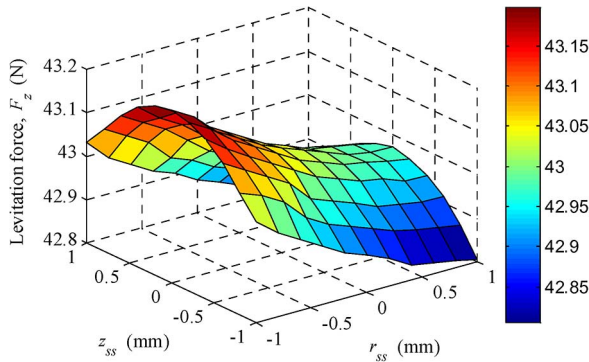


Fig. 22. Resultant levitation force distribution of the MLGC prototype.

dynamic force of ± 4 N is generated. The actuator constant, which is defined as F/i per turn, is about 0.125 N/A.

VI. CONCLUSION

In this paper, a novel cylindrical MLGC using Halbach magnet array has been presented; the compensator possesses excellent characteristics, such as compact size, low stiffness, and vacuum compatibility. Passive magnetic compensation and active control are effectively integrated into the MLGC. On the one hand, the main part of the moving mass of the fine stage is compensated by passive magnetic levitation. Therefore, the power loss and heat of the active Lorentz coils are reduced, which is beneficial for improving the system-level positioning accuracy. On the other hand, better vibration isolation can be achieved due to the low vertical stiffness.

Based on the equivalent current method, the analytical model of the MLGC is established. However, the traditional analytical model cannot accurately describe the characteristics of the MLGC because several key factors are neglected. The first factor is that the relative permeability is assumed to be 1.0. The second factor is the neglect of the difference between the attractive force and the repulsive force. Therefore, we proposed a simple and effective method to improve the accuracy of the analytical model based on a single-point magnetostatic field simulation. Additionally, the improved semianalytical model shows a good agreement with the finite-element model. It provides an important reference for the design and optimization of the MLGC. A prototype of the MLGC was manufactured and

tested. The measured levitation force matches well with the theoretical analysis and the finite-element simulation. Moreover, the vertical stiffness of the MLGC within the effective range of motion is quite low, which is beneficial when applied to the high-precision positioning stage.

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